

ORIGINAL ARTICLE

Effect of *Bombax ceiba* L. on spermatogenesis, sexual behaviour and erectile function in male ratsC. Bhargava¹, M. Thakur¹ & S. K. Yadav²¹ Advance Group of Pharmacy Colleges, Naramau Kanpur, India;² Ravi Shankar College of Pharmacy, Bhopal, India**Keywords***Bombax ceiba*—semalmusli—sexual performance—testosterone**Correspondence**

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Summary

A number of herbal drugs are advocated in the traditional Ayurvedic literature for the improvement of overall sexual function. Young roots of *Bombax ceiba* Linn. (Fam. Bombacaceae) [correction added after online publication 1 August 2011: the family of *Bombax ceiba* was incorrectly mentioned as Orchidaceae. It has been corrected to Bombacaceae] also known as Semal Musli are used traditionally in Indian subcontinent as sexual stimulant. Its juice is considered nutritive and restorative tonic. Lyophilised aqueous extract of roots was studied for effect on sexual behaviour and spermatogenesis in male albino rats. Administration of 100 mg Kg⁻¹ body weight of aqueous extract influenced the five parameters evaluated *in vivo*. Sexual behaviour analysis in the presence of a female rat, serum testosterone level, anabolic effects, epididymal sperm count and seminal fructose level were the parameters evaluated. In *B. ceiba* extract-treated animals, a gain in body and sexual organ weights was observed. Mount, intromission and ejaculation frequencies were significantly improved ($P < 0.05$). An increase in serum testosterone levels was also observed, but it was not statistically significant ($P > 0.05$). Seminal fructose content and epididymal sperm count were significantly improved as well. Penile erection index was also higher compared to control group animals. Hesitation time was significantly reduced ($P < 0.01$), and copulatory rate was doubled in treated animals compared with control group animals.

Introduction

The term sexual dysfunction equivocally implies for a number of sexual disorders ranging from erectile dysfunction to impotence. A longer life expectancy and the increasing prevalence of degenerative diseases as well as the increase in injuries and stress associated with industrialised lifestyles have been a cause of sexual dysfunction and impotency especially in males (Shah, 2002). Both medical and surgical modalities are available for treating sexual dysfunction. However, despite the increasing availability of effective conventional medical treatments, plant-derived and herbal remedies continue to be a popular alternative for men and women seeking to improve their sex life. There are number of such herbs that have been recognised in Ayurveda and Unani literature for improving sexual problems and diseases (Bhattarai, 1996; Suter, 1998).

Bombax ceiba Linn. Fam. Bombacaceae is a medicinal tree used in India for a number of health benefits and is

a part of the lexicon in Ayurvedic and Unani system of medicine (Khare, 2004). Young roots of *B. ceiba* find use in traditional Ayurvedic literature as 'Semal Musli' which has been designated as Ayurvedic Rasayana. Charaka and Suhruta Samhita mention the use of this herb alone as well as in combination for the treatment of leucorrhoea, morbid thirst and as geriatric tonic (Puri, 2003). It is also prescribed for sexual debility, attenuated semen and spermatorrhea (Anonymous, 1998; Khouri *et al.*, 2006). It is with these frames of references in mind the present study was carried out to evaluate the fabled claim of utilisation of this herb as a sexual rejuvenator and tonic.

Materials and methods**Animal stock**

Wistar strain albino rats of either sex weighing 180–195 g were fed on standard pellet diet and water *ad libitum*.

The animals were housed at room temperature ($24 \pm 2^\circ\text{C}$) on a reversed day-night cycle (06:00 to 18:00 h). All the animals were allowed to acclimatise in the test cage 7 days prior to experimentation. Observational and behavioural analyses were performed in a wooden chamber with a glass wall ($80 \times 30 \times 50$ cm) under diffused red light in the dark phase of the light dark cycle. The chamber had a special small opening at the side for introducing the female as stimulus. The video-recorded data were subjected to analysis using free-ware version of Etholog v 2.2.5[®] E.B. Ottoni (Sau Paulo) (Ottoni, 2000). All the animal experimentations were carried out after prior permission from the institutional ethical committee of Advance Institute of Biotech and Paramedical Sciences, India. The guidelines of Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), India were adhered to during the whole experimentation.

Preparation of extracts

Dried young roots of *B. ceiba* were procured from local market in Kanpur, India and identified at the Department of Pharmacy, Advance Institute of Biotechnology and Paramedical Sciences, Kanpur (India). Voucher specimen for the plant has been submitted (CSB-BC-09). Extraction was carried out as per the previously reported methodology (Thakur & Dixit, 2006).

Preparation of test samples

Aqueous extract was reconstituted in 0.2% PVP solution and administered orally using a metal canula. Testosterone suspension (Aquaviron; Nicholas Piramal India Ltd., Mumbai, India) purchased from the market was suspended in arachis oil and administered intramuscularly.

Treatment

Wistar strain male albino rats in a group of six each were taken for all the studies and marked.

Following the pre-test, the successful males were randomly assigned to three groups. Group I was administered vehicle only and served as control. Group II was administered with a daily dose of aqueous extract 100 mg Kg^{-1} b. w. orally. Group III was given 0.5 mg Kg^{-1} b. w. of suspension of testosterone propionate in arachis oil twice weekly intramuscularly (Saxena & Dixit, 1987).

For sexual behaviour analysis, 18 female rats were rendered oestrus by adopting the methodology reported by Agmo (2001).

Studies performed

Sexual behaviour

Determination of attraction towards sexually receptive female was carried out using the methods reported by Ang & Ngai, (2001) modified by Thakur & Dixit, (2006). On 14th and 28th day of treatment, female rat was placed in a cage that had a wooden barrier of 15 cm separating male and female compartments which could be passed by a motivated male rat.

The hesitation time was determined as per the methodology of Thakur *et al.*, (2009a).

Past acclimatisation of 5 min, sexually receptive female rat was introduced silently from one side of the chamber as stimulus. The whole pattern was digitally recorded, and observations for various parameters viz. Mount latency (ML) were calculated as the time from the introduction of female to the occurrence of first mount. Intromission latency (IL) was considered the time for first intromission after introduction of female in the cage. Intromission frequency started with a mount and results in vaginal penetration, usually the male starts licking the erect genital past intromission. Intromission ratio was determined by dividing the number of intromissions by the sum of number of mounts and number of intromissions (Ferguson *et al.*, 1970; Benelli *et al.*, 1997; Agmo, 2003; Thakur *et al.*, 2009b). Post-ejaculatory interval (PEI) was calculated as time from ejaculation until next intromission. Mount frequency was considered as the total number of mounts within 30 min.

Penile erection index

Determination of penile erection in noncontact position was calculated by the methodology reported by Matsumoto *et al.*, (1997). In brief, male rat was placed in a transparent cabin divided in half by two sheets of plastic fibre mesh preventing contact but allowing auditory, visual and olfactory stimuli. Ventral as well as lateral viewing and recording of the whole experimentation was facilitated by appropriately placing a mirror. After a 5-min adaptation period, placing an oestrous female on the other side of the cage started the test. Special attention was given to proper cleaning of the cage to avoid any carryover olfactory signals. Number of erections were recorded and tabulated. Erection in rats was marked by the visibility of the penis out of its sheath. Grooming of penis is another indicator of penile erection in rats and recorded as well. Penile erection index (PI) was calculated as per previously reported methodology (Islam *et al.*, 1991).

Serum testosterone levels

On day 0 and day 28, blood samples of the rats were collected from the retro-orbital plexus, and testosterone levels were determined by HPLC as per the methodology reported by Bonilla-Jaime *et al.*, (2006).

Epididymal sperm count

After the determination of body and organ weights, the determination of epididymal sperm counts was carried out according to the methodology reported by Wang, (2003).

Corrected seminal fructose content

Seminal vesicles along with adherent coagulating gland were removed carefully and stored at -25°C until further use. The tissue was then allowed to attain room temperature and further subjected to fructose content determination (Gonzales & Villena, 2001). Corrected seminal fructose content, which correlates with sperm motility, was calculated as follows:

Corrected seminal fructose content = $(\log_{10} \text{ sperm count}) \times (\text{seminal fructose concentration})$.

Statistical analysis

Results are reported as mean $\pm$ SE. The treated groups were compared to control by ANOVA following Dunnet's test. Significance level was set at $P < 0.5$ and confidence level at 95%. Statistical analysis was carried out using Instat v 2.1 software residing in a Pentium IV processor run on Windows XP[®].

Results

Sexual behaviour

Treatment with lyophilised aqueous extract of *B. ceiba* extract had a pronounced effect on sexual behaviour of the animals. A significant increase in mount, intromission and ejaculation frequency was observed in case of *B. ceiba*-treated animals ($P < 0.05$). Testosterone treatment caused an increase in frequencies although the effect was not significant when compared to control group. The sexual behaviour of the animals was influenced after treatment with aqueous extract of *B. ceiba*. The mount latency, intromission latency and post-ejaculatory latency were significantly reduced in extract-treated groups. Mount latency time that is an indicator of physical exhaustion during a sexual act was reduced by 38% in *B. ceiba* treated and 34% in testosterone-treated group as

compared to control group. Intromission (IL) and PEL time was reduced by 34% in *B. ceiba*-treated group ($P < 0.05$), whereas only 17% reduction was observed in testosterone-treated group. Reduced IL and PEL have been correlated with invigoration of endocrine system thereby, resulting in enhanced sexual performance and motivation (Figs 1 and 2).

Penile erection index (PEI) indicates the involvement of nitrous oxide (NO)-based mechanism whereby the conversion of L- arginine to L-citrulline takes place releasing NO which diffuses in the cavernosal tissues resulting in retention of blood in the vessels resulting in erection (Bancroft, 2005). An increase in PEI was observed in treated groups; *B. ceiba*-treated animals were best in this regard ($P < 0.01$) while testosterone-treated animals were the next to follow ($P < 0.05$) when compared to control group animals (Table 1).

Serum testosterone levels

Testosterone levels were affected by treatment with *B. ceiba* extract ($3.1 \pm 0.2 \text{ ng ml}^{-1}$), although the levels were not significantly higher than the control group animals ($1.6 \pm 0.1 \text{ ng ml}^{-1}$). In contrast, there was a higher serum testosterone level in case of testosterone-injected animals ($2.4 \pm 0.1 \text{ ng ml}^{-1}$) ($P < 0.5$).

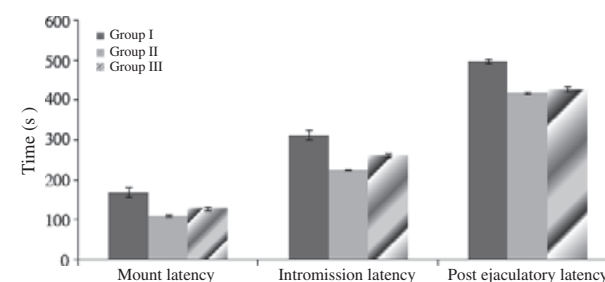


Fig. 1 Effect of treatment with *B. ceiba* extract on sexual behavior (latency time).

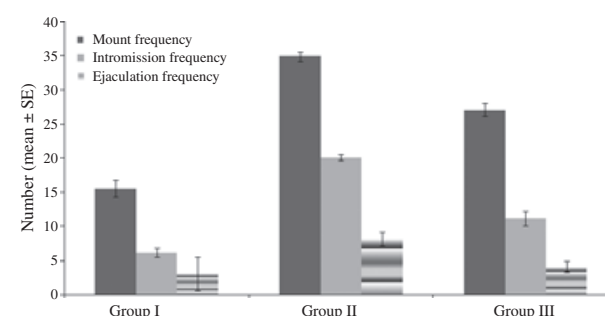


Fig. 2 Effect of treatment with *B. ceiba* extract on sexual behavior (frequencies).

Table 1 Effect of drug treatment on orientation and sexual behaviour parameters after 28 days of treatment

Parameters	Group I	Group II	Group III
Penile erection index (PEI)	21.2 ± 0.9	62.1 ± 1.1**	51.2 ± 1.3**
Hesitation time (mean time ± SE in s)	330 ± 8	102 ± 3**	138 ± 7*
Cumulative score for attraction towards female	22 ± 2	81 ± 1**	76 ± 1**
Serum testosterone levels in ng ml ⁻¹	1.6 ± 0.1	3.1 ± 0.2*	2.4 ± 0.1*
Corrected seminal fructose content (mg g ⁻¹ of seminal vesicles)	5.6 ± 0.1	7.9 ± 0.1**	4.7 ± 0.3

* $P < 0.05$ considered significant, ** $P < 0.01$ considered very significant.

Group I—Control (treated with vehicle only).

Group II—*Bombax ceiba* aqueous extract (100 mg kg⁻¹ b. w.) p. o. daily and exposed to heat damage.

Group III—Testosterone in arachis oil (0.5 mg kg⁻¹ b. w.) twice weekly by intramuscular route.

Epididymal sperm count and seminal fructose content

The corrected seminal fructose concentration in mg g⁻¹ of seminal vesicle was found to be 5.6 ± 0.1 in control group animals. For *B. ceiba* extract-treated group, the value was 7.9 ± 0.1, the value was lower for testosterone-treated group (4.7 ± 0.3) than the control.

Discussion

Semal Musli has been indicated for use as rejuvenative tonic and sexual capacity enhancers in various traditional systems of medicine practiced throughout the world. It is considered an important aphrodisiac plant in Ayurvedic, Unani and literature and is employed to enhance performance as well as to increase vigour and vitality. Although used frequently in various formulations, there is a considerable lacuna as far as scientific information backing the claim is concerned. The observed anabolic activity evidenced by gain in body and organ weights indicates testosterone intervention of the drug extracts (Bancroft, 2005). It is quiet likely that the components of the extracts are helpful in ensuring the better availability of testosterone to gonads incrementing spermatogenesis.

The drug treatment affected sexual behaviour of the animals positively. There was an increased sexual intent in male rats after *B. ceiba* treatment. Increased attraction towards female and reduced hesitation time provides ample evidence that the presence of a positive stimuli results in an enhanced vigour and involvement of animal in sexual manoeuvre. A very high PEI observed in the treated group clearly indicates that the herb may be

acting via nitrous oxide-based mechanism as well, which is chiefly responsible for an increased blood flow to penis resulting in improved erectile function (Bonilla-Jaime *et al.*, 2006).

Increased level of testosterone in adult males results in an increased sexual desire and arousability. An increase observed in plasma testosterone level could also be correlated with an increased penile rigidity even after the stimulus, which is a sexually receptive female, is removed, increased testosterone level results in nonstimulus-bound penile rigidity for a longer duration (Thakur *et al.*, 2011). Increase in testosterone level has been associated with a moderate but significant increase in sexual desire as well. Peripheral responses in nervous system are dependent on an increased testosterone level although central nervous system (CNS) is independent. Therefore, an improved serum testosterone level after administration of extracts could be considered one of the contributing factors responsible for an overall incremented sexual performance in treated groups. Aetiology of erectile dysfunction has also been correlated with low serum testosterone levels (Wang, 2003). Enhanced levels of testosterone after treatment could be foreseen as a possibility for usage of the herb in treatment of erectile dysfunction as well. It is worth mentioning that increased testosterone level alone is not the sole contributors in enhanced sexual performance as the testosterone-treated groups with higher circulating testosterone level were not superior to extract-treated group in overall sexual performance evaluation. It clearly indicates that a superior performance of plant extracts could be due to the interplay of other mechanisms along with enhanced hormonal activity (Thakur *et al.*, 2009a).

Bombax ceiba is an effective herb in improving the functionality of sexual organ. Present study validates the function of the plant as aphrodisiac and sexual performance improving agent. The results also corroborate the folkloric claim that the plant is capable of being nominated as herbal cure for sexual dysfunction and impotency. Further studies underpinning the chemical characterisation of the herb and their effect on testicular functions could provide substantial evidence in coming future.

References

- Agmo A (2001) Male rat sexual behavior. *Brain Res Protocol* 1:203–209.
- Agmo A (2003) Lack of opioid or dopaminergic effects on unconditioned sexual incentive motivation in male rats. *Behav Neurosci* 117:55–68.
- Ang HH, Ngai TH (2001) Aphrodisiac evaluation in non-copulator male rats after chronic administration of *Eurycoma longifolia* Jack. *Fund Clin Pharmacol* 15:265–268.

- Anonymous (1998) Hamdard Pharmacopoeia of Eastern Medicine. Vol. 1. Hamdard Publication, New Delhi.
- Bancroft J (2005) The endocrinology of sexual arousal. *J Endocrinol* 186:411–427.
- Benelli A, Arletti R, Poggioli R, Cavazutti E, Mameli M, Bertolini A (1997) Effect of potassium channel modulator on male sexual behavior. *Life Sci* 60:263–267.
- Bhattarai NK (1996) Some endangered medicinal plants of Nepal. In: Supplement to Cultivation and Utilization of Medicinal Plants. Handa SS, Kapur VK (eds). Regional Research Laboratory, Jammu, pp. 676–677.
- Bonilla-Jaime H, Vázquez-Palacios G, Arteaga-Silva M, Retana-Márquez S (2006) Hormonal responses to different sexually related conditions in male rats. *Horm Behav* 49:376–382.
- Ferguson H, Hemnriksen S, Cohen H, Mitchell G, Barcha J, Dement W (1970) Hypersexuality' and behavioral changes in cats caused by administration of p-chlorophenylalanine. *Science* 168:499–501.
- Gonzales GF, Villena A (2001) True corrected seminal fructose level: a better marker of the function of seminal vesicles in infertile men. *Int J Androl* 24:255–260.
- Islam MW, Tariq M, Ageel AM, Al-Said MS, Al-Yhya AM (1991) Effect of *Salvia hematodes* on sexual behavior of male rats. *J Ethnopharmacol* 33:67–72.
- Khare CP (2004) Encyclopedia of Indian Medicinal Plants: Rational Western Therapy Ayurvedic and Other Traditional Usage, Botany. Vol. 1. Springer Verlag, Berlin.
- Khoury NA, Nawasreh M, Alhussain SM, Alkofahi AS (2006) Effects of orchids (*Orchis anatolica*) on reproductive function and fertility in adult male mice. *Reprod Med Biol* 5:269–276.
- Matsumoto T, Kondo Y, Sachs BD, Yamanouchi K (1997) Effect of p- chlorophenylalanine on reflexive and noncontact penile erections in male rats. *Physiol Behav* 61:165–168.
- Ottoni EB (2000) EthoLog 2.2 - a tool for the transcription and timing of behavior observation sessions. *Behav Res Meth Inst Comput* 32:446–449.
- Puri HS (2003) Ayurvedic Herbs for Longevity and Rejuvenation. Taylor and Francis, London.
- Saxena S, Dixit VK (1987) Role of total alkaloids of *Mucuna pruriens* Baker. *Indian J Natl Prod* 3:3–7.
- Shah J (2002) Erectile dysfunction through the ages. *Br J Urol Int* 90:433–441.
- Suter S (1998) Impotence: evaluation and treatment in general practice – what is reliable. *Ther Umsch* 55:357–360.
- Thakur M, Dixit VK (2006) Effect of *Chlorophytum borivilianum* on androgenic and sexual behavior in male rats. *Indian Drugs* 43:300–306.
- Thakur M, Bhargava S, Chauhan NS, Dixit VK (2009a) A Comparative study on aphrodisiac activity of some Ayurvedic herbs in male albino rats. *Arch Sex Behav* 38:1009–1015.
- Thakur M, Bhargava S, Dixit VK (2009b) Effect of *Asparagus racemosus* on sexual dysfunction in hyperglycemic male rats. *Pharm Biol* 47:390–395.
- Thakur M, Thompson D, Connellan P, Deseo MA, Morris C, Dixit VK (2011) Improvement of penile erection, sperm count and seminal fructose levels *in vivo* and nitric oxide release *in vitro* by ayurvedic herbs. *Andrologia* 43:273–277.
- Wang Y (2003) Epididymal sperm count. *Curr Prot Toxicol* 14:16.16.11–16.16.15.